Chapter 4

Universals and Taxonomic Structures

A central concern of this thesis is to construct a philosophical and cognitive ontology that can be used as a foundation for conceptual modeling languages. Moreover, we aim at formally characterizing the elements constituting this ontology. Finally, we intend to demonstrate the usefulness of the ontological categories and theories proposed in addressing recurrent problems in the practice of conceptual modeling.

The construction of the foundation ontology proposed in this thesis is organized in four complementary chapters, namely, chapters 4 to 7. In this chapter, we aim at providing ontological foundations for the philosophical categories of *universals* and *individuals*, which are represented in conceptual modeling by the constructs of *Types (classes, classifiers)* and their *instances*, respectively.

Types are fundamental for conceptual modeling, being represented in all major conceptual modeling languages (e.g., OO classes, EER Entity types, LINGO and OWL concepts). In general, monadic types used in structural conceptual models stand for universals whose instances are *substantials*. The precise notions of *substantial* adopted in this thesis will be formally defined in chapter 6. However, for now, an intuitive understanding of this term will suffice. The term, as used here, is akin to what is sometimes named *thing* (Bunge, 1977), *endurant* (Masolo et al., 2003a), or *continuant* (van Leeuwen, 1991) in the philosophical literature. Intuitively, it is similar to what is termed *object*, in the colloquial use of the latter term. Substantials are entities that persist in time while keeping their identity (as opposed to *events* such as a kiss, a business process or a birthday party). Examples include physical and social persisting entities of everyday experience such as balls, rocks, students, the North Sea and Queen Beatrix.

In the practice of conceptual modeling, a set of primitives is often used to represent distinctions in different types of substantial universals (Type, Role, State, Mixin, among others). However, there is still a lack of methodological support for helping the user of the language to decide how to represent elements that denote universal properties in a given domain (viz. Person, Student, Red Thing, Physical Thing, Deceased Person, Customer) and, hence, modeling choices are often made in an ad hoc manner. Likewise is the judgment of what are the admissible relations between these modeling constructs. Finally, an inspection of the literature shows that there is still much debate on the meaning of these categories (Wieringa & de Jong & Spruit, 1995; Bock & Odell, 1998; Steimann, 2000b; Evermann & Wand, 2001b).

In this chapter, we propose a philosophically and psychologically well-founded theory of substantial universals for conceptual modeling. The ontological distinctions and postulates proposed by this theory are presented in section 4.1.

In section 4.2, the ontological distinctions countenanced by the theory are organized in a typology of universals, together with a number of constraints on how the elements in this typology can be combined to form taxonomic structures. This typology and associated constraints are further used to derive a *modeling profile* for conceptual modeling, along with a set of methodological guidelines that govern its use. Still in section 4.2, we demonstrate the usefulness of the theory and derived profile proposed to evaluate and improve the conceptual quality of class hierarchies and concept taxonomies, and to solve some recurrent problems in the practice of conceptual modeling.

In section 4.3, we present a number of empirical research efforts carried out in cognitive psychology that provide evidence supporting the proposed theory of universals.

In section 4.4, we elaborate on two different (albeit complementary) systems of modal logics designed to formally characterize the distinctions and constraints proposed by the theory. The section also discusses how these systems address the limitations of classical (unrestricted extensional) modal logics in that respect.

Section 4.5 discusses related work and demonstrates how the ontological distinctions proposed in this chapter are compatible but richer than those found in the conceptual modeling literature hitherto.

Finally, section 4.6 elaborates on some final considerations.

4.1 A Theory of Universal Types: Philosophical and Psychological Foundations

In (van Leeuwen, 1991), Jacques van Leeuwen shows an important grammatical difference occurring in natural languages between common nouns (CNs) on one side and arbitrary general terms (adjectives, verbs, mass nouns, etc…) on the other. Common nouns have the singular feature that they can be combined with determiners and serve as argument for predication in sentences such as:

(i) (exactly) five mice were in the kitchen last night;
(ii) the mouse which has eaten the cheese, has been in turn eaten by the cat.

In other words, if we have the patterns *(exactly) five X…* and *the Y which is Z…*, only the substitution of X, Y, Z by CNs will produce sentences that are grammatical. To verify that, we can try the substitution by the adjective Red in the sentence (i): *(exactly) five red were in the kitchen last night.* A request to "count the red in this room" cannot receive a definite answer: Should a red shirt be counted as one or should the shirt, the two sleeves, and two pockets be counted separately so that we have five reds? The problem in this case is not that one would not know how to finish the counting but that one would not know how to start, since arbitrarily many *subparts of a red thing are still red.*

It is important to emphasize that red here is not used as a CN, i.e., as a synonym for red color, which is a nominalization of an adjective and denotes a particular shade of red. This reading would make a sentence such as *"exactly 256 greys exist in the Windows color palette"* grammatically viable. In fact, in order to play the same role as a CN, general terms must be nominalized, which implies a shift to the category of common nouns (e.g., whiteness, the fall of Jack or a Bucket of water).

The explanation for this feature unique of CNs lies on the function that determinates (demonstratives and quantifiers) play in noun phrases, which is to determine a certain range on individuals. Both reference and quantification requires that the thing (or things) which are referred or which form the domain of quantification are determinate individuals, i.e., their conditions for individuation and identity must be determinate. In other words, if it is not determinate how to count Xs or how to identify the X that is the same as Y, the sentences in the patterns (i) and (ii) do not express determinate propositions, i.e., propositions with definite truth values.

According to (van Leeuwen, 1991), this syntactic distinction between the two linguistic categories reflects a semantical and ontological one, and the distinction between the grammatical categories of CNs and arbitrary general terms can be explained in terms of the ontological categories of *Sortal* and *Characterizing universals* (Strawson, 1959), which are roughly their ontological counterparts. Whilst the latter supply only a *principle of application* for the individuals they collect, the former supply both a principle of application and a *principle of identity*. A principle of application is that in accordance with which we judge whether a general term applies to a particular (e.g., whether something is a Person, a Dog, a Chair or a Student). A principle of identity supports the judgment whether two particulars are the same, i.e., in which circumstances the identity relation holds.

In (Mcnamara, 1986), cognitive psychologist John Macnamara, investigates the role of sortal concepts in cognition and provides a comprehensive theory for explaining the process that a child undergoes when learning proper nouns and common nouns. He proposes the following example: suppose a little boy (Tom), who is about to learn the meaning of a proper name for his puppy. When presented to the word "Spot", Tom has to decide what it refers to. A demonstrative such as "that" will not suffice to determinate the bearer of the proper name. How to decide that "that", which changes all its perceptual properties is still *Spot*? In other words, which changes can Spot suffer and still be the same? As Macnamara (among others) shows, answers to these questions are only possible if *Spot* is taken to be a proper name for an individual, which is an instance of a Sortal universal. The principles of identity supplied by the Sortals are essential to judge the validity of all identity statements. For example, if for an instance of the sortal *Statue* loosing a piece will not alter the identity of the object, the same does not hold for an instance of *Lump of Clay*.

The statement that we can only make identity and quantification statements in relation to a Sortal amounts to one of the best-supported theories in the philosophy of language, namely, that the identity of an individual can only be traced in connection with a Sortal Universal, which provides a *principle of individuation* and *identity* to the particulars it collects (Mcnamara, 1986, 1994; Gupta, 1980; Lowe, 1989; van Leeuwen, 1991). The position advocated in this chapter affirms an equivalent stance for a theory of conceptual modeling. We defend that among the conceptual modeling counterparts of general terms (*types*) only constructs that represent sortals can provide a principle of identity and individuation for its instances. As a consequence, a principle that represents the junction of

Quine's dicto *"no entity without identity"* (Quine, 1969) with the position defended in this section *"no identity without a Sortal"* can be postulated:

Postulate 4.1: Every individual in a conceptual model (CM) of the domain must be an instance of a conceptual modeling type (CM-Type) representing a sortal.

As argued by Kripke (Kripke, 1982), a proper name is a rigid designator, i.e. it refers to the same individual in all possible situations, factual or counterfactual. For instance, it refers to the individual Mick Jagger both now (when he is the lead singer of Rolling Stones and 62 years old) and in the past (when he was the boy Mike Philip living in Kent, England). Moreover, it refers to the same individual in counterfactual situations such as the one in which he decided to continue in the London School of Economics and has never pursued a musical career. We would like to say that the boy Mike Philip is identical with the man Mick Jagger that he latter became. However, as pointed out by Wiggins (Wiggins, 2001) and Perry (Perry, 1970), statements of identity only make sense if both referents are of the same type. Thus, we could not say that a certain Boy is the same Boy as a certain Man since the latter is not a Boy (and vice-versa). However, as Putnam put it, when a man *x* points to a boy in a picture and says "I am that boy", the pronoun "I" in question is typed not by Man but by a supertype of Man and Boy (namely, Person), which embraces *x*'s entire existence (Putnam, 1994). A generalization of this idea amounts to a thesis, proposed by Wiggins, named thesis D (Wiggins, 2001): If an individual falls under two sortals in the course of its history there must be exactly one ultimate sortal of which both sortals are specializations. (Griffin, 1977) elaborates Wiggins' thesis D in terms of two correlated principles:

a) The Restriction Principle: if an individual falls under two distinct sortals F and F' in the course of its history then there is at least one sortal of which F and F' are both specializations.

b) The Uniqueness Principle: if an individual falls under two distinct sortals F and F' in the course of its history then there is at most one *ultimate sortal* of which F and F' are both specializations. A sortal F is ultimate if there is no other sortal F' distinct from F which F specializes.

It is not the case that two incompatible principles of identity could apply to the same individual *x*, otherwise *x* will not be a viable entity (determinate particular) (van Leeuwen, 1991). For instance, suppose an individual *x* which is an instance of both *Statue* and *Lump of clay*. Now, the answer to the question whether loosing a piece will alter the identity of *x* is indeterminate, since each of the two principles of identity that *x* obeys imply a different answer. As a consequence, we can say that if two sortals F and F' intersect (i.e., have common individuals in their extension), the principles of identity contained in them must be equivalent. Moreover, F and F' cannot supply a principle of identity for *x*, since both sortals apply to x only contingently, and a principle of identity must be used to identify *x* in all possible worlds. Therefore, there must be a sortal G that supplies the principle of identity carried by F and F'. This proves the restriction principle. The uniqueness of the ultimate sortal G can be argued as follows: (i) G is a sortal, since it supplies a principle of identity for all the things in its extension; (ii) if it restricts a sortal H then, since H cannot supply an incompatible principle of identity, H either is: equivalent to G (i.e., does supply the same principle of identity) and therefore should be ultimate, or does not supply a principle of identity for the particulars in its extension (see text on dispersive classifiers below). This proves the uniqueness principle. The unique ultimate sortal G that supplies the principle of identity for its instances is named a *substance sortal*.

As a consequence of thesis D, we derive a second postulate:

Postulate 4.2: An individual represented in a conceptual model of the domain must instantiate exactly one CM-Type representing an ultimate Substance Sortal.

In the example above, the sortal Person is the *unique* substance sortal that defines the validity of the claim that Mick Jagger is the same as Mike Philip or, in other words, that Mike Philip persists through changes in height, weight, age, residence, etc., as the same individual. Person can only be the sortal that supports the proper name Mick Jagger in all possible situations because it applies necessarily to the individual referred by the proper name, i.e., instances of Person cannot cease to be so without ceasing to exist. As a consequence, the extension of a substance sortal is world invariant. This meta-property of universals is named *Modal Constancy* (Gupta, 1980) or *rigidity* (Guarino & Welty, 2002b) and is formally stated as follows:

Definition 4.1 (Extension functions): Let W be a non-empty set of possible worlds and let $w \in W$ be a specific world. The extension function $ext_w(G)$ maps a universal G to the set of its instances in world w. The extension function ext(G) provides a mapping to the set of instances of the universal G that exist in all possible worlds, such that

1. $\text{ext}(G) = \bigcup_{w \in W} \text{ext}_w(G)$

■

Definition 4.2 (Specialization relation): Let F and G be two universals such that F is a specialization of G. Then, for all $w \in W$ we have that

2. $\text{ext}_w(F) \subseteq \text{ext}_w(G)$

■

Definition 4.3 (Rigid Universal): A universal G is rigid (or modally constant) iff for any $w,w' \in W$

3. $\text{ext}_w(G) = \text{ext}_{w'}(G)$

■

Putting definitions 4.1 and 4.3 together, we have that for any rigid universal G the following is true

4. $\text{ext}(G) = \text{ext}_w(G)$, for all $w \in W$

A rigid universal is one that applies to its instances necessarily, i.e., in every possible world. Every substance sortal G is a rigid universal. Examples of *non-rigid* Sortals include universals such as *Boy* and *Adult Man* in the example discussed above, but also *Student*, *Employee*, *Caterpillar* and *Butterfly*, *Philosopher*, *Writer*, *Alive* and *Deceased*. Actually, these examples of sortals are not only non-rigid, but they are *anti-rigid*. Non-rigidity is the simple logical negation of rigidity, i.e., a universal is non-rigid if it does not apply necessarily to at least one of its instances. In contrast, a universal is anti-rigid if it does not apply necessarily to all its instances. An example of a non-rigid universal is *Seatable*. Suppose the instances of Seatable include a particular chair *x* and a particular crate *c*. In this case, whilst *x* instantiates Seatable necessarily, this is not the case for *c*: a crate can cease to be steady to afford sitting but still be the same crate.

Non-rigidity and anti-rigidity of universals are defined formally in the sequel:

Definition 4.4 (Non-Rigid Universal): A universal G is non-rigid iff for a $w \in W$

5. There is an x such that $x \in ext_w(G)$, and there is a $w' \in W$ such that $x \notin ext_{w'}(G)$

■

Definition 4.5 (Anti-Rigid Universal): A universal G is anti-rigid iff for any $w \in W$

6. For every x such that $x \in ext_w(G)$, there is a $w' \in W$ such that $x \notin ext_{w'}(G)$

■

Notice that non-rigidity constitutes a much weaker constraint than what is imposed by anti-rigidity, i.e. anti-rigidity is a sort of non-rigidity. In (Guarino & Welty, 2002b), a universal is named *semi-rigid* iff it is non-rigid but not anti-rigid.

Sortals that possibly apply to an individual only during a certain phase of its existence are named *phased-sortals* in (Wiggins, 2001). Contrary to substance sortals, phased-sortals are anti-rigid universals. For example, for an individual John instance of Student, we can easily imagine John moving in and out of the Student type, while being the same individual, i.e. without loosing his identity. Moreover, for every instance x of Student in a world w, there is another world w' in which x is not an instance of Student. Finally, as a consequence of the Restriction Principle, we have that for every phased-sortal PS that applies to an individual, there is a substance sortal S of which PS is a specialization. Formally a phased-sortal can be defined as follows:

Definition 4.6 (Phased-Sortal): Let PS be a universal and let S be a *substance sortal* specialized (restricted by) PS. Now, let

7. $ext_w(\sim PS) = ext_w(S) \setminus ext_w(PS)$

be the complement of the extension of PS in world w. In this formula, the symbol \ represents the set theoretical operation of set difference. The universal PS is a *phased-sortal* iff for all worlds $w \in W$, there is a $w' \in W$ such that

8. $ext_w(PS) \cap ext_{w'}(\sim PS) \neq \emptyset$

■

Putting (2), (4) and (6) together we can derive another postulate:

Postulate 4.3: A CM-Type representing a rigid universal cannot specialize (restrict) a CM-Type representing an anti-rigid one.

Proof: Take an arbitrary rigid universal G which specializes an anti-rigid universal F. Let $\{a,b,c,d\}$ and $\{a,b\}$ be the extension of F and G in world w, respectively. By (6), there is a world w' in which $a \notin ext_{w'}(F)$. By (4), however, $ext_w(G) = ext_{w'}(G)$ and, thus, $a \in ext_{w'}(G)$. By (2), $ext_{w'}(G) \subseteq ext_{w'}(F)$ and, ergo, $a \in ext_{w'}(F)$. We then have that $a \notin ext_{w'}(F)$ and $a \in ext_{w'}(F)$, which is a contradiction. We therefore conclude that a rigid universal cannot specialize an anti-rigid one.

□

If PS is a phased-sortal and S is the substance sortal specialized by PS, there is a specialization condition φ such that *x* is an instance of PS iff *x* is an instance S that satisfies φ (van Leeuwen, 1991). A further clarification on the different types of specialization conditions allows us to distinguish between two different types of phased-sortals which are of great importance to the practice of conceptual modeling, namely, *phases* and *roles*.

Phases (also named *dynamic subclasses* in Wieringa & de Jonge & Spruit, 1995) or states (Bock & Odell, 1998) constitute possible stages in the history of a substance sortal. Examples include: (a) Alive and Deceased: as possible stages of a Person; (b) Catterpillar and Butterfly of a Lepidopteran; (c) Town and Metropolis of a City; (d) Boy, Male Teenager and Adult Male of a Male Person. *Universals representing phases constitute a partition of the substance sortal they specialize*. For example, if ‹Alive, Deceased› is a *phase-partition* of a substance sortal Person then for every world w, every Person x is either and instance of Alive or of Deceased but not of both. Moreover, if x is an instance of Alive in world *w* then there is world *w*' such that x is not an instance of Alive in *w*', which in this case, implies that x is an instance of Deceased in w'.

This can be generalized as follows: Let $\langle P_1 \ldots P_n \rangle$ be a phase-partition that restricts the sortal S. Then we have that for all $w \in W$:

9. $$ext_w(S) = \bigcup_{P_i \in \langle p_1 \ldots p_n \rangle} ext_w(P_i)$$

and for all $P_i, P_j \in \langle P_1 \ldots P_n \rangle$ (with $i \neq j$) we have that

10. $$ext_w(P_i) \cap ext_w(P_j) = \emptyset$$

Finally, it is always possible (in the modal sense) for an instance *x* of S to become an instance of each P_i, i.e., for any $P_i \in \langle P_1 \ldots P_n \rangle$ which restricts S, and for any instance *x* such that $x \in ext_w(S)$, there is a world w' ∈ W such that $x \in ext_{w'}(P_i)$. This is equivalent of stating that for any $P_i \in \langle P_1 \ldots P_n \rangle$ the following holds

11. $ext(S) = ext(P_i)$

Contrary to phases, roles do not necessarily form a partition of substance sortals. Moreover, they differ from phases with respect to the specialization condition φ. For a phase P, φ represents a condition that depends solely on intrinsic properties of P. For instance, one might say that if Mick Jagger is a Living Person then he is a Person who has the property of being alive or, if Spot is a Puppy then it is a Dog who has the property of being less than an year old. For a role R, conversely, φ depends on extrinsic (relational) properties of R. For example, one might say that if John is a Student then John is a Person who is enrolled in some educational institution or that, if Peter is a Customer then Peter is a Person who buys a Product y from a Supplier z. In other words, an entity plays a role in a certain context, demarcated by its relation with other entities. In general, we can state the following: Let R be a role that specializes a sortal S (named the *allowed type* for R (Bock & Odell, 1998)) and φr be a binary relation defined between R and the universal D on which R is *externally dependent of* (Welty & Guarino, 2001). For instance, $\varphi_{enrollment} \subseteq Student \times School$, $\varphi_{purchase\text{-}from} \subseteq Customer \times Supplier$ or $\varphi_{Marriage} \subseteq Husband \times Wife$. Moreover, let the domain of the relation φr in world *w* (Dom_w) be defined as follows: $Dom_w(\varphi_r) = \{x \mid <x,y> \in ext_w(\varphi_r)\}$. Then for all worlds $w \in W$ we have that

12. $ext_w(R) = Dom_w(\varphi_r)$

Relational properties are represented in this chapter in terms of plain (extensional) relations for simplicity only. In chapter 6, we present the complete treatment of relations and relational properties which is adopted in our foundational ontology. That is, in turn, used in chapter 7 to elaborate on a fuller characterization of roles. For the same reason of simplicity, we will thus not extend the representation of relations in the example above to the case of n-ary relations.

Although Frege argued at length that *"one cannot count without knowing what to count"* (Frege, 1934), in artificial logical languages inspired by him, natural language general terms such as common nouns, adjectives and verbs are treated uniformly as predicates. For instance, if we want to represent the sentence "there are tall men", in the fregean approach of classical logic we would write $\exists x\ (Man(x) \wedge Tall(x))$. This reading puts the count noun Man (which denotes a Sortal) on an equal logical footing with the predicate Tall. Moreover, in this formula, the variable x is interpreted into a "supposedly" universal kind Thing (or Entity). So, the natural language reading of the formula should be "there are things that have the property of being a man and the property of being tall". Since, by postulate 4.1, all individuals must be instances of a substance sortal we must conclude that Thing is a unique universal ultimate sortal that is able to supply a principle of identity for all elements that we consider in our universe of discourse. Moreover, by postulate 4.2, this principle of identity must be unique. Can that be the case?

In (Hirsch, 1982), Hirsch argues that concepts such as Thing, (Entity, Element, among others) are *dispersive*, i.e., they cover many concepts with different principles of identity. For instance, in the extension of Thing we might encounter an individual x that is a cow and an individual y that is a watch. Since the principles of identity for Cows and Watches are not the same, we conclude that Thing cannot supply a principle of identity for its instances. Otherwise, x and y would obey incompatible principles of identity and, thus, would not be determinate individuals. Therefore, as defended in (van Leeuwen, 1991; Gupta, 1980; Mcnamara, 1994; Hirsch, 1982), dispersive concepts do not denote sortals, despite that they are considered CNs in natural languages, and *therefore cannot have direct instances.* More than that, a principle of identity supplied by a substance sortal G is inherited by all universals that specialize G or, to put in another way, all subtypes of G carry the principle of identity supplied by G. Thus, all specializations of a sortal are themselves sortals, ergo,

Postulate 4.4: A CM-Type representing a dispersive universal cannot specialize a CM-Type representing a Sortal.

4.2 An Ontologically Well-Founded Profile for Conceptual Modeling Universals

We start this section by compiling the ontological distinctions proposed by the theory of section 4.1 in a *typology of substantial universals*. The elements constituting this typology are depicted in figure 4.1 below.

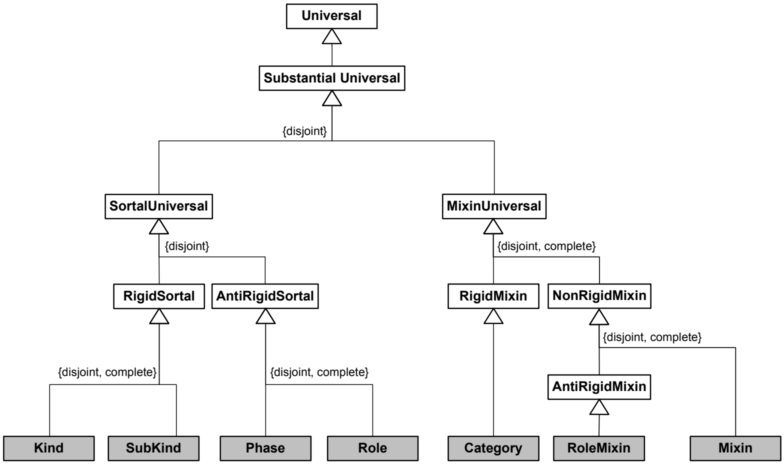


Figure 4-1 Ontological Distinctions in a Typology of Substantial Universals

The idea here is to use the ontology of universals depicted above to design a fragment of a conceptual (ontology) modeling language. This language should contain (as modeling primitives) constructs that represent the bottom-most specialization of the ontological categories proposed by this theory, i.e., the leaf nodes of the generalization tree of figure 4.1 (highlighted in grey).

We use the extension mechanisms of the Unified Modeling Language (UML) to illustrate these ideas by proposing a *profile* whose modeling elements represent each of the relevant distinctions made by the theory. It is important to emphasize, however, that UML is used here solely with the purpose of exemplification, due to its acceptance and practical relevance, and to the convenience offered by the built-in extension mechanisms of the language. A similar result could be achieved by extending, for instance, LINGO via its *theory inclusion* extension mechanisms (Falbo & Menezes & Rocha, 1998), or simply by proposing a new modeling language.

The UML built-in extension mechanisms allow one to modify the language elements to suite certain modeling needs. Extensions to the language can be performed in two different ways: (i) by specializing the UML metamodel (layer 2) to add new semantics to UML modeling elements; or (ii) by changing the so-called MOF model (layer 3) to add new elements to the UML metamodel. The former mechanism is named *lightweight extension* and the latter *heavyweight extension*. A coherent set of such extensions, defined accordingly to a specific purpose or domain, constitutes a *UML profile* (OMG, 2003b).

In a *representation mapping* (see chapter 2) from the ontology of figure 4.1 to the UML metamodel, we can map the category of *universal* to the UML meta-construct of a *Class*. In UML, a *"Class describes a set of Objects* *sharing a collection of Features, including Operations, Attributes and Methods, that are common to the set of Objects."* (OMG, 2001, p.32). *"The model is concerned with describing the intension of the class, that is, the rules that define it. The run-time execution provides its extension, that is, its instances."* (ibid., p.202). The concept underlying this construct, namely, the one of a *Type*, is one of the most common modeling concepts in conceptual modeling. For instance, it can be found practically in all object-modeling languages (e.g., OMT, Objectory, OML), semantic web languages (e.g., OWL, DAML+OIL), ontology modeling languages (e.g., LINGO) and information systems grammars (e.g., ER). For this reason, the principles and distinctions laid out in section apply to any of these conceptual modeling languages in which the substantial universal concept is represented.

If we continue this representation mapping, we realize that in UML (but also in all of the languages aforementioned) there are no modeling constructs that represent the ontological categories specializing *Substantial Universal* in figure 4.1. In other words, there are ontological concepts prescribed by the theory of section 4.1 that are not represented by any modeling construct in the language. A case of ***ontological incompleteness*** at the modeling language level (see section 2.2.4).

In order to remedy this problem, in the sequel, we propose a lightweight extension to UML that represents finer-grained distinctions between different types of classes. The proposed *profile* contains a set of *stereotyped* classes that support the design of ontologically well-founded conceptual models according to the theory proposed in this chapter. Moreover, the profile also contains a number of constraints (derived from the postulates of the theory) that restrict the way the modeling constructs can be related. The goal is to have a metamodel such that all syntactically correct specifications using the profile have logical models that are *intended world structures* of the conceptualizations they are supposed to represent (see definition 3.2 in chapter 3).

In summary, according to the ontological semantics given to the profile, the ontological interpretation of the class meta-construct is that of a *Universal*. Each of the stereotyped classes comprising the profile, thus, represents one of the leaf ontological categories specializing substantial universal in figure 4.1. In terms of the profile, this also means that all stereotyped classes have as the *base class* the UML Class meta-construct (OMG, 2003b).

In the subsections that follow, we elaborate in each of these distinctions. It is important to emphasize that the particular classes chosen to exemplify each of the proposed categories are used for illustration purposes only. For example, when stereotyping class *Person* as a *Kind* we are not advocating that Person should be in general considered as a kind in conceptual modeling.

Conversely, the intention is to make explicit the consequences of this modeling choice. The choice itself, however, is always left to the model designer.

4.2.1 Kinds and Subkinds

A UML class in this profile stereotyped as a « kind » represents a *substance sortal* that *supplies* a principle of identity for its instances. Kinds can be specialized in other *rigid* subtypes that inherit their supplied principle of identity named *subkinds*. For instance, if we take Person to be a kind then some of its subkinds could be Man and Woman. In general, the stereotype « subkind » can be omitted in conceptual specifications without loss of clarity.

Every object in a conceptual specification using this profile must be an instance of a Kind, directly or indirectly (postulate 4.1). Moreover, it cannot be an instance of more than one ultimate Kind (postulate 4.2). Figure 4.2 shows an excerpt of a conceptual specification that violates the second postulate (extracted from the CYC[28] upper-level ontology). Here, we assume that the kinds *Social Being* and *Group* supply different principles of identity. Moreover, it is considered that *Group* supplies an extensional principle of identity, i.e. two groups are the same iff they have the same members. This is generally incompatible with a principle supplied by *Social Being*: we can change the members of a company, football team or music band and still have the same social being. Moreover, the same group can form different social beings with different purposes. One should notice that if the particular referred by the proper name *The Beatles* would be an instance of both Kinds, it would not be a determinate object (an answer to the question whether it was still the same thing when Ringo Star replaced Pete Best, is both affirmative and negative!). Figure 4.3 shows a version of the specification of figure 4.2 that obeys the constraints of this profile. In this revised representation, we have explicit modeled the *substantial* The Beatles and the group of people that compose this individual in a given circumstance as distinct individuals.

[28] http://www.opencyc.org/

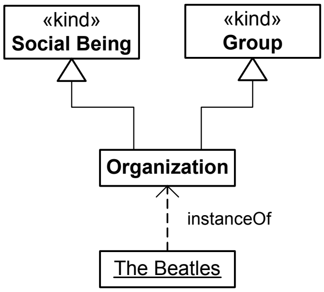


Figure 4-2 Example of an instance with conflicting principles of identity

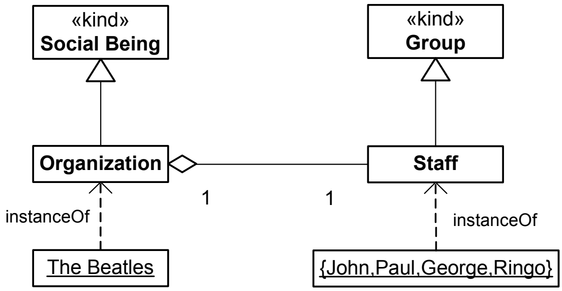


Figure 4-3 An ontologically correct version of the specification of figure 4-2

By postulate 4.3, rigid classes cannot be supertyped by anti-rigid ones. Therefore, kinds cannot appear in conceptual models as subtypes of phases, roles (see section 4.2.3), and role mixins (section 4.2.4).

4.2.2 Phases

UML classes stereotyped as « phase » in this profile represent the phased-sortals *phase*. Figure 4.4 depicts an example with the kind Person, its subkinds Man and Woman and the phases Child, Adolescent and Adult.

The classes connected to one single hollow arrowhead symbol in UML (concrete syntax for the subtyping relation) define a *generalization set* (OMG, 2003b). In the UML 2.0 metamodel, the classes that are member of a generalization are necessarily disjoint. However, by default, they do not form a partition (in the mathematical sense) of the restricted subclass. In order to represent that a set of subclasses form a partition of their direct common superclass, the corresponding generalization set must be decorated with the constraints {*disjoint, complete*}. In other words, the concept of a *type partition* is represented as a decorated generalization set in UML.

The kind person in figure 4.4 represents the substance sortal restricted by a *phase-partition* ⟨Child, Adolescent, Adult⟩. According to formulas (9) and (10) in section 4.1, the generalization set representing a phase-partition must indeed by a partition in the mathematical sense and, hence, the restricted superclass must always be defined as an *abstract class* (i.e., a class that cannot have direct instances). In the specification of figure 4.4, the subkinds Man and Woman also define a partition for the substance sortal Person. However, contrary to phases, subkinds do not have to be necessarily defined in a partition.

In UML, an abstract class is represented by a class with its name italicized.

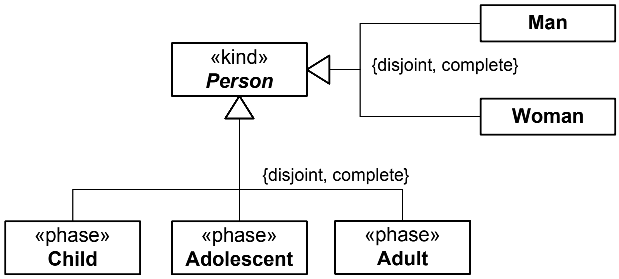


Figure 4-4 Two partitions of the same Kind Person: a subkind partition (Man, Woman) and a phase partition (Child, Adolescent, Adult)

4.2.3 Roles

UML classes stereotyped as « role » represent the phased-sortals *role*. Roles and Phases are anti-rigid universals and cannot appear in a conceptual model as a supertype of a Kind (postulate 4.3). However, sometimes subtyping is wrongly used in conceptual modeling to represent alternative *allowed types* that can fulfill a role. For instance, in figure 4.5, the intention of the model is to represent that customers are either persons or organizations. An analogous example is shown in figure 4.6. In general, being a customer is assumed to be a contingent property of person, i.e., there are possible worlds in which a Person is not a customer but still the same person. Likewise, a participant can stop participating in a Forum without ceasing to exist. In summary, if the universals represented in figures 4.5 and 4.6 are ascribed the semantics just discussed, these specifications are ontologically incorrect representations according to the theory postulated in section 4.1.

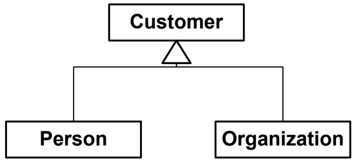


Figure 4-5 Problems on modeling Roles and their *allowed types*

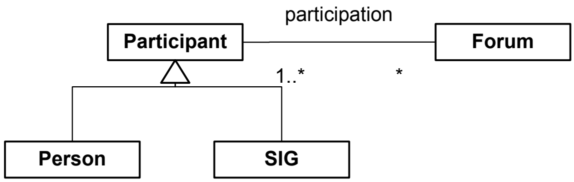


Figure 4-6 Mistaken cardinality specification for Roles

In a series of papers (Steimann, 2000a, 2000b, 2001), Steimann discusses the difficulties of specifying allowed types for Roles that can be played by instances of disjoint types such as the roles *Customer* and *Participant* in figures 4.5 and 4.6, respectively. As a conclusion, the author claims that the solution to this problem lies in the separation of role and type hierarchies. This solution not only departs from the traditional use of the concepts of role and type in the practice of conceptual modeling but, in particular, it leads to a radical revision of the UML metamodel (a heavyweight extension). In chapter 7 of this thesis, while discussing several problems w.r.t. role modeling found in the literature of conceptual modeling, we show that this claim is not warranted. To support our argument and propose a solution to the problem of role modeling with disjoint allowed types, we propose a design pattern based on the profile presented in this section. The solution presented has a smaller impact to UML than the one proposed by the author, since it does not demand heavyweight extensions to the language. The *roles with disjoint allowed types design pattern* proposed in chapter 7 can then be used as an ontologically correct solution to this recurrent modeling problem.

Figure 4.7 below depicts an ontologically correct version of figure 4.5 generated by the application of the design pattern aforementioned. We return to the discussion of this issue and, specifically, of this particular example in chapter 7. The notion of role mixin in figure 4.7 is discussed in section 4.2.4 below.

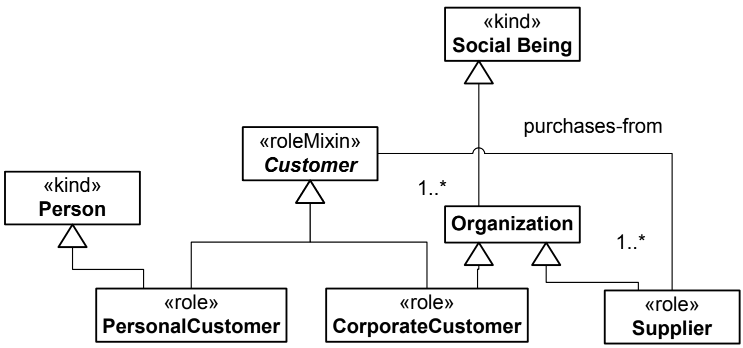


Figure 4-7 An ontologically correct version of the specification of figure 4-5 by using the profile and the *roles with disjoint allowed types* design patterns

Figure 4.6 contains yet another conceptual problem. In this specification, a participant can take part in zero-to-many forums. It is common in Database and Object-Oriented Design to use a minimum cardinality equal to zero to express that in a certain state of the system, for example, an object of type Participant is not related to any object of type Forum. However, from a conceptual viewpoint, the involvement in this relation is part of the very definition of the role type. In this example, the association *participation* is a specialization (restriction) condition (see section 4.1), which is part of the content of the concept Participant. In other words, a *Participant* is a *Person* or *SIG* that takes part in a *Forum*. As a consequence of formula (7), the following constraint must hold for classes stereotyped as « role »:

Let X be a class stereotyped as « role » and r be an association representing X's restriction condition. Then the minimum cardinality of X.r must be at least 1 (#X.r ≥ 1).

This constraint is elaborated in chapter 8 of this thesis after our treatment of relations is presented and formally characterized in chapter 6.

4.2.4 Mixins

Conceptual modeling types classified as Mixins represent the dispersive universals discussed in section 4.1. Mixin types are perceived to be of great importance in structuring the specification of conceptual models (Jacobson & Booch & Rumbaugh, 1998; Booch, 1994; Welty & Guarino, 2001). They can represent top-most types such as Thing, Entity, Element (discussed in section 4.1) but also concepts such as *RationalEntity*, which represent an abstraction of properties that are common to multiple disjoint types (figure 4.8). In this case, the mixin *RationalEntity* can be judged to represent an essential property that is common to all its instances and it is itself a rigid type. We use the stereotype «category» to represent a rigid mixin that subsumes different kinds.

In contrast, some mixins are anti-rigid and represent abstractions of common properties of roles. These types are stereotyped as «roleMixin» and represent dependent anti-rigid non-sortals. Examples of role mixins include the so-called *formal roles* (Welty & Guarino, ibid.) such as *whole* and *part* and *initiatior* and *responder*. However, role mixins are more general than formal roles, representing any abstraction of common contingent properties of multiple disjoint roles. An example of a role mixin is depicted in figure 4.7. Further examples are discussed in chapter 7 of this thesis.

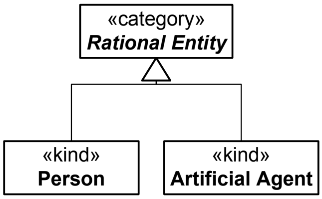


Figure 4-8 Example of a *category*, i.e., a rigid mixin.

Moreover, some mixins represent properties that are essential to some of its instances and accidental to others. As discussed in section 4.1, this meta-property is named non-rigidity (a weaker constraint than anti-rigidity). An example is the mixin *Seatable* (figure 4.9), which represents a property that can be considered essential to the kinds Chair and Stool but accidental to Crate, Paper Box or Rock. We use the stereotype « mixin » (without further qualification) to represent *non-rigid* non-sortals.

Finally, by postulate 4.4, we have that mixins cannot appear in a conceptual model as subclasses of kinds, phases or roles. Moreover, due to postulate 4.3, rigid mixins (categories) cannot be subsumed by anti-rigid ones, i.e., by role mixins. Finally, since they cannot have direct instances, a mixin must always be depicted as an abstract class in a UML conceptual specification.

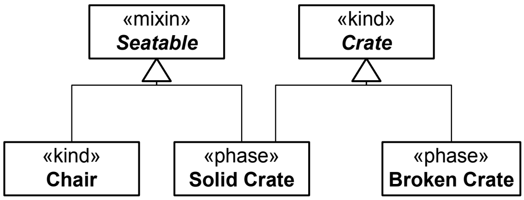


Figure 4-9 Example of a semi-rigid mixin

Table 4.1 below summarizes the profile proposed in this section.

Table 4-1 Summary of the proposed modeling profile for the conceptual modeling representation of substantial universals

Stereotype	Constraints
	RIGID SORTALS
« kind »	supertype is **not** a member of {« subkind », « phase », « role », « roleMixin »}
« subkind »	supertype is **not** a member of {« phase », « role », « roleMixin »} For every subkind SK there is a unique kind K such that K is a supertype of SK
	ANTI-RIGID SORTALS
« phase »	Always defined as part of partition. For every Phase P there is a unique Kind K such that K is a supertype of P
« role »	Let X be a class stereotyped as « role » and R be an association representing X's restriction condition. Then, $\#X.R \geq 1$ For every Role X there is a unique Kind K such that K is a supertype of X

	NON-SORTALS
« category »	supertype is **not** a member of {« kind », « subkind », « phase », « role », « roleMixin »} Always defined as an abstract class
« roleMixin »	supertype is **not** a member of {« kind », « subkind », « phase », « role »}. Let X be a class stereotyped as « roleMixin » and R be an association representing X's restriction condition. Then, #X.R ≥ 1 Always defined as an abstract class
« mixin »	supertype is **not** a member of {« kind », « subkind », « phase », « role », « roleMixin »} Always defined as an abstract class

4.3 Psychological Evidence

The postulates presented in section 4.1 are represented in a list of psychological claims proposed by cognitive psychologist John Mcnamara in (Mcnamara, 1994). The proposed psychological claims are related to the cognitive interpretation of linguistic expressions. In that article, Mcnamara defends the position that there is a logic underlying the fact that we can understand certain propositions, and the proposed set of psychological claims points to a class of logics that take cognizance of this fact. This position, which is also supported by, for instance, (Mcnamara, 1986; Xu, 2004; Lidz & Waxman & Freedman, 2003) is analogous to the one advocated by Chomsky in defense of his notion of a Universal Grammar (Chomsky, 1965, 1980, 1986). Chomsky is the proponent of the theory that states that the reason why we can learn a natural language is due to the existence of a *mental language*, a *linguistic competence* that is nature-supplied, uniform across individuals and complete in each one. According to this view, there is a close fit between the mind's linguistic properties and properties of natural languages and, hence, natural languages have the properties they do because they can be recognized and manipulated by infants without the meta-linguistic support that is available to second-language learners. Therefore, for Chomsky, a grammar for a particular language is *descriptively adequate* if it correctly describes its object, namely the linguistic intuition of the native speaker.

In the same spirit, a number of cognitive scientists (see, for example, Mcnamara 1986, 1994), have proposed a theory of *logical competence* based on the notion of a *Language of Though* (Fodor, 1975). For him, the reason we can learn the *meaning* of natural language symbols for categories, individuals and their properties without the meta-linguistic support available to second language learners lies on the mapping between the language specific symbols for these categories onto a system of categories already existing in the language of though, i.e., a *cognitive ontology*.

In this section, we discuss some empirical evidence that support many of the points defended throughout this chapter. This evidence results from a number of psychological experiments in the areas of cognitive psychology, which has been developed with the aim of investigating categorical development and logical competence in infants since a pre-language age of 3-4 months.

Firstly, many laboratory results provide evidence that infants in the early age of 3-4 months are already able to form categories (e.g., Eimas & Quinn, 1994). (Cohen & Younger, 1983; Mcnamara, 1982), for instance, provide evidence that children are able to classify objects into categories for which they have no natural-language symbols. The reason that categorization appears so early in cognitive development is related to its fundamental relevance to cognition. As (Markman 1989, p. 11) puts it: *"Categorization . . . is a means of simplifying the environment, of reducing the load on memory, and of helping us to store and retrieve information efficiently".* Without concepts, mental life would be chaotic. If we perceived each entity as unique, we would be overwhelmed by the sheer diversity of what we experience and unable to remember more than a minute fraction of what we encounter. Furthermore, if each individual entity needed a distinct name, our language would be staggeringly complex and communication virtually impossible. In contrast, if you know nothing about a novel object but are told it is an instance of X, you can infer that the object has all or many properties that Xs have (Smith & Medin, 1981).

Nonetheless, in order to able the learn what are the properties that we can expect instances of X to have, another ability, namely the ability to (re)identity instances of X must be present. If all of an object's properties were immediately manifest to the child upon every encounter there would be no need to learn and remember what these properties were. However, carrying knowledge of substances becomes necessary since most of a substance's properties are not manifest but hidden from us most of the time (Milikan, 1998). For example, different encounters with Felix will reveal different properties about the individual Felix and about the kind Cat. However, for this to happen, the child must be able to: (i) recognize that Felix *is a Cat*; (ii) recognize that the individual that can jump from the table to the TV set *is the same* as the one that can be fed milk; (iii) recognize that Tom *is also a Cat*, thus, he can also be fed milk. In summary, both principles of application and identity are fundamental in our construal of the environment.

A number of experiments provide evidence for the existence of an individuation and indentity system in infants by the age they begin to form categories. Researchers such as (Xu & Carey, 1996; Spelke et. al., 1995; Moore, Borton, & Darby, 1978) provide evidence that by the early age 3-4 months old infants have criteria for deciding whether an object is the same one as a previously seen object. (Xu & Carey, ibid.), for instance, provide evidence that until about nine months of age, infants rely on a unique principle of individuation and identity for all objects, which is supplied by the type *Physical Object*. This notion of physical object is synonymous to *maximally connected physical object whose parts move along together in a spatiotemporal continuous path*. As defended by the authors, in this sense, physical object is indeed a sortal since any identity and individuation statements involving two physical objects is determinate. They term this system of individuation an *object-based system*. These results show that not only does spatiotemporal discontinuity lead to a representation of two distinct objects, but also that spatiotemporal continuity leads to a representation of a single, persisting object. Other laboratories have also replicated this basic finding using somewhat different procedures (e.g., Aguiar & Baillargeon, 1999).

Thus, even young infants have some criteria for establishing representations of distinct objects. These first criteria are spatiotemporal in nature, including generalizations such as: (i) objects travel on spatiotemporally connected paths; (ii) two objects cannot occupy the same space at the same time, and (iii) one object cannot be in two places at the same time (Xu, 2004).

Xu and Baker (Xu & Baker, 2003) address the question whether infants can use perceptual property information for object individuation, i.e., if non-sortal categories can support the judgment of individuation statements. The results show that, for 10-month infants, perceptual properties are at best only used to confirm the application of the principle of identity which is first supplied by the sortal *Physical Object*. Moreover, these results show that *spatiotemporal evidence for a single object changing properties overrides perceptual property information* (see also Xu & Carey, 1996). (Xu & Carey & Quint, 2004) also shows that 12-months infants are not able to use non-sortal categories alone to support the judgment of individuation statements. This view is also supported by (Milikan, 1998) who calls the attention for the fact that children come to appreciate separable dimensions, such as color, shape, and size only after a considerable period in which "holistic similarities" dominate their attention.

Results from (Xu & Carey & Quint, 2004) show that between 9 to 12 months of age a second system of individuation emerges in infant's cognition. This is named a *kind-based individuation system* and operates independently of the object-based attention system. In general, by the early age of 12 months, infants have already developed the multiple-sortal system which is used by adults to judge individuation and identity statements. Moreover, (Xu, 2004) shows that: (i) representations of object kinds can override strong spatiotemporal evidence for a single object; (ii) perceptual property information is always treated as kind-relative. As remarked by the author, *"during this period, infants' worldview undergoes fundamental changes: They begin with a world populated with objects...By the end of the first year of life, they begin to conceptualize a world populated with sortal-kinds... In this new world, objects are thought of not as 'qua object' but rather 'qua dog' or 'qua table'"*. Other experiments such as those conducted by Bonatti et al. (2002), Waxman and Markow (1995) and Booth and Waxman (2003) corroborate with these findings and support the claim that around 12-months infants are sensitive to the distinction between sortals and arbitrary general terms, which are represented differently and used differently in individuation tasks.

(Xu, 2004) also suggests that it is not a coincidence that along with acquiring their first words, infants begin to develop their kind-based system of individuation. The bulk of a child's first words are concrete nouns, including proper names and names for sortal universals (Milikan, 1998). For example, (Mcnamara, 1972) shows that children learn common nouns (the linguistic counterparts of sortal universals) before they learn predicates such as verbs and adjectives (counterparts of characterizing universals). (Gentner, 1982) presents evidence that this finding holds cross-culturally for children learning German, Kaluli, Japanese, Mandarin Chinese, Turkish and English. In summary, in early stages of language learning, children are more likely to pick words for sortals than of other kinds. Although perceived, attributes and events are construed, individuated and conceptualized under the dependence of a sortal (Mcnamara, 1986, p.145).

(Katz & Baker & Mcnamara, 1974) and (Mcnamara, 1982) provide evidence that children younger than 17-month-old are able to distinguish proper names by coordinating the notions of individual and kind. According to these findings, children are able to judge that individuals of some kinds, but not of others, are likely to be the bearers of a proper name. Together with the results from (Gelman & Taylor, 1984), these findings provide strong indications that under certain circumstances children are led to take some words as proper names (namely, when applied to individuals of familiar kinds) and in others to take them as sortals (when applied to individuals of unfamiliar kinds). (Mcnamara, 1986) advocates that this evidence is an obvious suggestion that children have the appropriate sortals to support the learning of proper names in some language. In addition to that, when reporting on a number of observations extracted from the linguistic record he kept of his son, Mcnamara remarks that his son's use of proper names behaved like rigid designators from the start (Mcnamara, 1986).

According to (Rosch et. al, 1976; Markman & Hutchinson, 1984), when learning a word for a kind of object, basic-level sortals (*substance sortals*) are the ones that occur to children. For example, when creating categories, children attend to similarities among dogs before subclassifying them and before they attend to properties dogs share with other animals. (Mcnamara, 1986, p.147) strongly argues that substance sortals hold a psychologically salient and privileged position compared to other types of rigid sortals, and that children's perceptual systems seem to be especially tuned to identify substance sortals. This position finds evidence in the results of (Anglin, 1977; Mervis & Crisafi, 1982).

In summary, a substantial number of psychological experiments confirm the philosophy of language thesis that individuation and identity judgment can only take place with the support of a sortal universal. Both systems of individuation and identity that are employed by human cognition are sortal-based. Humans start with a principle of identity afforded by the unique sortal physical object and, in a later developmental phase, undergo a cognitive shift to a system that employs a multitude of principles of identity supplied by different substance sortals. In both systems, perceptual property information is secondary and can be overridden. (Xu, 2004) defends the position that this developmental process makes good sense in terms of learnability, since it starts the child on the solid ground of a concept of object that seems to be inate (Spelke, 1990) and it allows the child to work with these individuated objects and with the help of language, ultimately develop a new ontology of sortal-kinds. In addition, as defended by (Milikan, 1998), the primacy of substance sortals also makes good sense in evolutionary terms since, from the standpoint of an organism that wishes to learn, the focus should be on constructing categorizations that are *essential*, since they are the ones that supply knowledge that affords the most useful and reliable inferences.

4.4 Formal Characterization

In this section we present two complementary systems of *intensional sortal modal logics* (intensional modal logics with quantification restricted by sortal terms). These systems differ mainly in the nature of the entities they admit to the domain of quantification. However, more importantly, they share the objective of making explicit the distinction between sortals and mixin universals (neglected in classical modal logics) in the sense that only sortals can carry a principle of persistence and trans-world identity for their instances. Moreover, they also incorporate in their semantics, the constraints that qualify the different types of sortals universals recognized by the theory presented in section 4.1.

4.4.1 Quantifying over Momentary States

In the use of the term *individuals* that we have been employing in this chapter, individuals are special types of endurants, i.e., entities that exist in time while keeping their identity. Examples including ordinary *objects* of everyday experience such as: a person, a student, a house, a dog, among others. However, in a UML Object Diagram (Instance Specification), an UML instance *"represents an entity at a point in time (a snapshot)"* (OMG, 2003c, p.59). In the language presented in this section, we explore this idea by taking the primitive elements of quantification in the system of intensional modal logics proposed to represent states of instances in given time boundaries, or to put it simply, snapshots of ordinary substantial individuals.

The following example illustrates some of these ideas. Suppose that there is an individual person referred by the proper name John. As discussed in section 4.1, proper names for substantials refer rigidly and, hence, if we say that John weights 80kg at t_1 but 68kg at t_2 we are in the two cases referring to the same individual, namely the particular John. Now, let x_1 and x_2 be snapshots representing the projection of John at time boundaries t_1 and t_2, respectively. The truth of the statements *overweight(John,t_1)* and *overweight(John,t_2)* depends only on whether overweight applies to the states x_1 or x_2, respectively. In other words, the judgment if an individual *i* is an instance of a universal G (e.g., overweight) in world w depends only whether the principle of application carried by G applies to the state of *i* in w.

In a computational system, the identity of x_1 and x_2 is guaranteed by the presence of some artificial identification scheme (oids, primary keys, surrogates). However, from an ontological perspective, how can one determine that, despite of possibly significant dissimilarities, x_1 and x_2 are states of the same particular John? As argued throughout the chapter, this is done via a principle of trans-world identity and, in particular, a principle of persistence supplied by the substance sortal Person, of which John is an instance.

In classical (extensional) modal logics, no distinction is made between different types of universals. Universals are represented as predicates in the language that divide the world (at each circumstance) into two classes of elements: those that fall under them and those that do not. This principle determines the extension of each universal at each circumstance. Classical (one-place) predicates, i.e., as functions from worlds to sets of individuals, properly represent the *principles of application* that are carried by all universals but fail to represent the *principles of identity* which are unique of sortals. Equivalently, they treat all objects as obeying the same principle of identity.

This difference is made explicit in the language L_1 defined in this section in the following way:

– The intention of the proper name John is represented by an *individual concept J*, i.e., a function that maps to a snapshop x_i of John in each possible world w. The notion of individual concepts, first introduced by Leibniz, refers to a singleton property that only holds for one individual. For instance, the property of being Mick Jagger has a single instance, namely the individual person Mick Jagger;

– Sortal universals, such as Person, are represented as *intensional properties*, which are functions from possible worlds to sets of individual concepts. For instance, for the sortal Person there is a function ℓ that maps every world w to a set of individual concepts (including *J*). An individual *x* is a Person in world w iff there is an individual concept $k \in \ell(w)$ such that $k(w) = x$;

– Individual Concepts represent the principle of identity supplied by the universal Person such that if $J(w) = x_1$ and $J(w') = x_2$ then we say that x_1 in w is the same Person as x_2 in w', or in general: for all individuals x,y representing snapshots of an individual C of type T we say that x in w is the same T as y in w' iff C is in the extension of T and $C(w) = x$ and $C(w') = y$;

– Whilst the principle of identity is represented by sortal determined individual concepts that trace individuals from world to world, the principle of application considers individuals only at a specific world. For instance, John is overweight in word w iff *overweight(J(w),w)* is true.

In summary, in the language L_1 presented below, the primitive elements in the domains of quantification are momentary states of substantial individuals, not the individuals themselves. Ordinary substantial individuals are instead represented by individual concepts.

The idea of representing objects of ordinary experience by individual concepts is similar to the solution presented in (Heller & Herre, 2004) in which individual concepts for substantial individuals are named *abstract substances* or *persistents*. The notion of momentary state of individuals adopted here is similar to that of *presentials* in (Heller & Herre, ibid.).

In the sequel, we formally define the syntax and semantics of L_1.

Syntax of L_1

Let L_1 be a language of modal logics with identity with a vocabulary V = (K,B,A,P,T) where: (a) T is a set of individual constants; (b) P is a non-empty set n-ary predicates; (c) A is a set of anti-rigid sortal types; (d) B is a set of subkinds; (e) K is a non-empty set of kinds (substance sortal type); (f) $R = K \cup B$ is named the set of rigid sortal types and the set $C = R \cup A$, the set of sortal types; The alphabet of L_1 contains the traditional operators: $=$ (equality), $\neg$ (negation), $\rightarrow$ (implication), $\forall$(universal quantification), $\Box$ (necessity). The notions of term, sortal and formula are define as follows:

Definition 4.7

(1) all individual constants and variables are *terms*;
(2) All sortal types belong to the category of *Sortal Types;*
(3) If s and t are terms, then $s = t$ is an *atomic formula*;
(4) If P is a n-place predicate and $t_1 \ldots t_n$ are terms, then $P(t_1,..,t_n)$ is an *atomic formula*;
(5) If A and B are formulas, then so are $\neg A$, $\Box A$, $(A \rightarrow B)$;
(6) If S is sortal classifier, x is a variable and A is a formula, then $(\forall S,x)A$ is a formula.

■

The symbols $\exists$ (existential quantification), $\wedge$ (conjunction), $\vee$ (disjunction), $\Diamond$ (possibility) and $\leftrightarrow$ are defined as usual:

Definition 4.8

(7) $(A \wedge B) =_{def} \neg(A \rightarrow \neg B)$;
(8) $(A \vee B) =_{def} ((A \rightarrow B) \rightarrow B)$;
(9) $(A \leftrightarrow B) =_{def} (A \rightarrow B) \wedge (B \rightarrow A)$
(10) $\Diamond A =_{def} \neg\Box\neg A$
(11) $(\exists S,x)\ A =_{def} \neg(\forall S,x)\ \neg A$
(12) $(\exists! S,x)\ A =_{def} (\exists S,y)(\forall S,x)\ (A \leftrightarrow (x = y))$

■

In L_1, all quantification is restricted by Sortals. The quantification restricted in this way makes explicit what is only implicit in standard predicate logics. As previously discussed, suppose we want to state the following proposition: (a) *There are red tasty apples*. In classical predicate logic we would write down a *logical* formula such as (b) $\exists$x (apple(x) $\wedge$ tasty(x) $\wedge$ red(x)). In an ontological reading, (b) states that "there are things which are red, tasty and apple". The theory proposed section 4.1 rejects that we can conceptually grasp an individual under a general concept such as Thing or Entity or, what is almost the same, that a logic (or conceptual modeling language) should presupposed the notion of a *bare particular*. Moreover, it states that only a sortal (e.g., Apple) can carry a principle of identity for the individuals it collects, a property that is absent in attributions such as Red and Tasty. For this reason, a logical system, when used to represent a formalization of conceptual models, should not presuppose that the representations of natural general terms such as Apple, Tasty and Red stand in the same logical footing. For this reason, (a) should be represented as ($\exists$Apple,x) (tasty(x) $\wedge$ red(x)) in which the sortal binding the variable x is the one responsible for carrying its principle of identity.

In L_1, sortal classifiers are never used in a predicative position. Therefore, if $S \in C$ is a sortal type, the predicate s(x) (in lowercase) is a meta-linguistic abbreviation according to the following definition.

Definition 4.9

$$s(t) =_{def} (\exists S,x)\ (x = t)$$

■

According to this definition, the sentence "John is a Man" is actually better rendered as "John is identical to a Man". In opposition, in the sentence "John is Tall", the copula represents the "is" of predication, which denotes a relation of mere equivalence.

Semantics of L_1

Definition 4.10 (model structure)

A model structure for L_1 is defined as an ordered couple $\langle \mathcal{W}, \mathcal{D} \rangle$ where: (i) W is a non-empty set of possible worlds; (ii) L_1 adopts a *varying domain frame* (Fitting & Mendelsonh, 1998) and, thus, instead of a set, $\mathcal{D}$ is a function that assigns to each member of $\mathcal{W}$ a non-empty set of elements.

■

Given a model structure $\mathcal{M}$ $(= \langle \mathcal{W}, \mathcal{D} \rangle)$, the intention of individual constant can be represented by an *individual concept*, i.e., a function i that assigns to each world $w \in \mathcal{W}$, an individual in $\mathcal{D}(w)$. Formally we have that

Definition 4.11 (individual concept)

Let $\mathcal{M} = \langle \mathcal{W}, \mathcal{D} \rangle$ and $\mathcal{U} = \bigcup_{w \in W} D(w)$.

An individual concept i in $\mathcal{M}$ is function from $\mathcal{W}$ into $\mathcal{U}$, such that $i(w) \in \mathcal{D}(w)$ in all worlds. For a given model structure $\mathcal{M}$ we define I as a set of individual concepts defined for that structure.

■

The intention of an n-place predicate is defined (as usual) as an n-ary property, i.e., a function that assigns to each world $w \in \mathcal{W}$ a set of n-tuples. If a tuple $\langle d_1 \ldots d_n \rangle$ belongs to the representation of a predicate at world w, then $d_1 \ldots d_n$ stand in w in the relation expressed by the predicate.

Definition 4,12 (property)

An n-ary property (n>0) in $\mathcal{M}$ is a function $\mathcal{P}$ from $\mathcal{W}$ into $\wp(\mathcal{D}(w))^n$, i.e., if $\langle d_1 \ldots d_n \rangle \in \mathcal{P}$(w), then $d_1 \ldots d_n \in \mathcal{D}(w)$.

■

The intension of sortal classifiers is defined in such a way that both the principles of application and identity are represented. This is done by what is named in (Gupta, 1980) *sorts*, i.e., *separated intensional properties*.

Definition 4.13 (sort)

Let $\mathcal{M} = \langle \mathcal{W}, \mathcal{D} \rangle$ be a model structure. An *intensional property* in $\mathcal{M}$ is a function ℓ from $\mathcal{W}$ into the powerset of individual concepts in $\mathcal{M}$ (i.e., $\wp(I)$).

An intensional property assigns to each world a set of individual concepts, and it can be used to represent the intention of a sortal type in the following way. Suppose that ℓ represents the intension of the sortal type $\mathcal{S}$ and that the individual concept i belongs to ℓ at world w, i.e., $i \in \ell(w)$. Then $i(w)$ is a $\mathcal{S}$ in w, and $i(w')$ is identical to $i(w)$ in w.

Let ℓ be an intensional property in $\mathcal{M}$, and let $\mathcal{L} = \bigcup_{w \in W} \ell(\mathrm{w})$.

Now, let i,j be two individual concepts such that $i,j \in \mathcal{L}$. We say that the intensional property ℓ is *separated* iff: if there is a world $w \in \mathcal{W}$ such that $i(w) = j(w)$ then, for all $w' \in \mathcal{W}$, $i(w') = j(w')$, i.e., $i = j$.

Finally, a *sort* in a model structure $\mathcal{M}$ is an intensional property which is separated.

■

The requirement of separation proposed in (Gupta, 1980) states, for example, that if two individual concepts for Person, say 007 and James Bond, apply to the same object in a world w then they apply necessarily to the same object. This prevents unlawful conceptualizations in which a substantial individual splits or in which two individuals can become one while maintaining the same identity.

Given a sort ℓ in $\mathcal{M}$ we designate by $\ell[w]$ the set of objects that fall under ℓ in w. Formally,

Definition 4.14:

$\ell[w] = \{d$: $d \in \mathcal{D}(w)$ and there is an individual concept $i \in \ell(w)$ such that $i(w) = d\}$.

■

Moreover, we define the set of objects in w that are *possibly* ℓ, i.e.,

Definition 4.15

$\ell\,|[w]| = \{d$: $d \in \mathcal{D}(w)$ and there is an individual concept $i \in \ell(w')$ such that $i(w') = d\}$.

■

We therefore are able to define the notion of *counterpart* relative to a sort ℓ.

Definition 4.16 (counterpart)

We say that d *in world* w *is the same* ℓ *as* d' *in* w' iff there is an individual concept i that belongs to ℓ at some world (i.e., there is a w'' such that $i \in \ell(w'')$) and $i(w) = d$ and $i(w') = d'$. The ℓ *counterpart* in w' of the individual d in w is the unique individual d' such that d' *in world* w' *is the same* ℓ *as* d *in* w.

■

Definition 4.17 (model)

A model in L_1 can then be defined as a triple $\langle \mathcal{W}, \mathcal{D}, \delta \rangle$ such that:

1. $\langle \mathcal{W}, \mathcal{D} \rangle$ is a model structure for L_1;

2. δ is an interpretation function assigning values to the non-logical constants of the language such that: it assigns an individual concept to each individual constant $c \in T$ of L_1; an n-ary property to each n-place predicate $p \in P$ of L_1; a sort to each sortal type $S \in C$ of L_1.

The interpretation function δ must also satisfy the following constraints.

3. If $S \in R$ then the sort ℓ assigned to S by δ must be such that: for all $w,w' \in \mathcal{W}$, $\ell(w) = \ell(w')$, i.e., all rigid sortals are world invariant (modally constant);

4. Let $S \in (B \cup A)$ be a subkind or an anti-rigid sortal type. Then, there is a kind $S' \in K$ such that, for all $w \in \mathcal{W}$, $\delta(S)(w) \subseteq \delta(S')(w)$;

5. Let $S, S' \in K$ be two Kinds and let ℓ and ℓ' be the two sorts assigned to S and S' by δ, respectively. Then we have that: there is a $w \in \mathcal{W}$ such that $\ell(w) \cap \ell'(w) \neq \emptyset$ iff $\ell = \ell'$, i.e., sorts representing kinds do not intersect unless they are identical. In other words, this restriction states that individuals belong to one single substance sortal, i.e., they obey one single principle of identity;

6. Let $S \in A$ be an anti-rigid sortal type. The sort ℓ assigned to S by δ must be such that: for all $w \in \mathcal{W}$, and for all individual concepts $i \in \ell(w)$, there is a world $w' \in \mathcal{W}$ such that $i \notin \ell(w')$;

7. Let $S, S' \in K$ be two Kinds and let ℓ and ℓ' be the two sorts assigned to S and S' by δ, respectively. Then we have that: there is a $w \in \mathcal{W}$ such that $\ell[w] \cap \ell'[w] \neq \emptyset$ iff $\ell = \ell'$. Differently from (5) above this restriction states that individual states of objects can only be referred by individual concepts of the same kind;

■

Finally, we are now in position to define an assignment for L_1.

Definition 4.18 (assignment)

An assignment for L_1 relative to a model $\langle \mathcal{W}, \mathcal{D}, \delta \rangle$ is a function that assigns to each variable of L_1 an ordered pair $\langle \ell, d \rangle$, where ℓ is a sort relative to the modal structure $\langle \mathcal{W}, \mathcal{D} \rangle$ and $d \in \mathcal{U} = \bigcup_{w \in W} D(w)$.

If a is an L_1 assignment then $a_o(x)$ is the object assigned to variable x by a and $a_S(x)$ is the sortal to which x is bound. Moreover, it is always the case that $a_o(x) \in a_S(x)\,|\,[w]\,|$ for all variables.

■

Definition 4.19

An assignment a' for L_1 is an ℓ variant of a at x in w iff:

1. a' is just like a except perhaps at x (abbreviated as $a' \sim_X a$),
2. $a'_s(x) = \ell$,
3. $a'_o(x) \in \ell[w]$.

■

Definition 4.20

The w' variant of an assignment a relative to w (abbreviated as $f(w',a,w)$) is the unique assignment a' that meets the following conditions:

(i) $a'_S(x) = a_S(x)$ at all variables x,

(ii) $a'_o(x)$ in w' is the $a_S(x)$ *counterpart* of $a_o(x)$ in w relative, at all variables x.

■

Definition 4.21

Finally, let α be an expression in L_1 and, let *the semantic value of* α *at world* w *in model* $\mathcal{M}$ *and relative to assignment* a be the value of the valuation function $v^w_{M,a}$.

With these definitions, we are able to define the semantics of L_1 as follows:

(a). If α is an individual constant or a sortal type, then $v^w_{M,a}(\alpha) = \delta(\alpha)(w)$.

(b). If α is variable, then $v^w_{M,a}(\alpha) = a_o(\alpha)$

(c). If α is an atomic formula $t_1=t_2$, then $v^w_{M,a}(\alpha) = T$ if $v^w_{M,a}(t_1) = v^w_{M,a}(t_2)$. Otherwise $v^w_{M,a}(\alpha) = F$.

(d). If α is an atomic formula $P(t_1 \ldots t_n)$, then $v^{w}_{M,a}(\alpha) = T$ if $\langle v^{w}_{M,a}(t_1)\ldots v^{w}_{M,a}(t_n)\rangle \in \delta(P)(w)$. Otherwise $v^{w}_{M,a}(\alpha) = F$.

(e). If α is the formula $\neg A$, then $v^{w}_{M,a}(\alpha) = T$ if $v^{w}_{M,a}(A) = F$. Otherwise $v^{w}_{M,a}(\alpha) = F$.

(f). If α is the formula $(A \rightarrow B)$, then $v^{w}_{M,a}(\alpha) = T$ if $v^{w}_{M,a}(A) = F$ or $v^{w}_{M,a}(B) = T$. Otherwise $v^{w}_{M,a}(\alpha) = F$.

(g). If α is the formula $(\forall S,x)A$, then $v^{w}_{M,a}(\alpha) = T$ if $v^{w}_{M,a'}(A) = T$ for all assignments *a'* which are $\delta(S)$ variants of *a* at x in *w*. Otherwise $v^{w}_{M,a}(\alpha) = F$.

(h). If α is the formula $\Box A$, then $v^{w}_{M,a}(\alpha) = T$ if $v^{w'}_{M,f(w',a,w)}(A) = T$ for all $w' \in \mathcal{W}$. Otherwise $v^{w}_{M,a}(\alpha) = F$.

■

The language L_1 has been proposed based on the first of the four systems introduced in (Gupta, 1980). Gupta, however, does not elaborate of different types of sortals. Consequently, restrictions from (3) to (7) posed to δ in definition 4.17 are simply not defined in his system. Restriction (7), in particular, would be rejected by Gupta, as a consequence of his contingent (or relative) view of identity. Note that restriction (7) implies (5) but not vice-versa.

It is widely accepted that any relation of identity must comply with the so-called Leibniz's law: if two individuals are identical then they are necessarily identical (van Leeuwen, 1991). Relativists, however, adopt the thesis that it is possible for two individuals to be identical in one circumstance but different in another. A familiar example, cited by Gupta, is that of a statue and a lump of clay. The argument proceeds as follows: Suppose that in world *w* we have a statue *st* of the Dalai Lama which is identical to the lump of clay *loc* this statue is made of. In w, *st* and *loc* have exactly the same properties (e.g., same shape, weight, color, temperature, etc.). Suppose now that in world w', a piece (e.g., the hand) is subtracted from *st*. If the subtracted piece is an inessential part of a statue then the statue *st'* that we have in *w'* is identical to *st*. In contrast, the lump of clay *loc'* which *st'* is made of is different from *loc*. In sum, we have in *w'* the same statue as in *w* but a different lump of clay.

In Gupta's system, without restriction (7), we have that for two individual concepts i and j such that $i(w) = j(w)$, it is possible that there is a world w' such that $i(w') \neq j(w')$. In other words, the formula (α) $((\exists \text{Statue},x)(x = dl) \wedge (\exists \text{LoC},y)(y = dl) \wedge \Diamond(x \neq y))$ is satisfiable.

Here, we reject this possibility for two reasons. Firstly, because we support the view that Leibniz's rule must hold for a relation to be considered a relation of identity. Otherwise, any equivalence relation such as *being instance of the same class* must be considered a relation of identity. Secondly, if Gupta's primitive elements are thought as momentary states, then (α) does not actually qualify as a statement of relative identity. It actually expresses that two objects can coincide (i.e., share the same state) in a world w but not in a different world w' (van Leeuwen, 1991). Notice that if restriction (7) is assumed, formula (α) is no longer satisfiable.

Proof: (a) if (x = dl) is true then there is an individual concept st of Statue that refer in the actual world w to the same entity d as dl; (b) if (y = dl) is true then there is an individual concept loc of LoC that refer in the actual world w to the same entity d as dl; (c) by transitivity of equality, st and loc refer to the same d in world w and, consequently, d is then both of kind Statue and of kind LoC in w; (e) due to (7), the intentions of Statue and LoC are identical; (f) finally, due to separation, st and loc must coincide in every world.

□

In L_1, we take the *multiplicationist* (Masolo et al., 2003a) position that *st* and *loc* do not actually share the same state in w in the strong sense. In opposition, we consider that the states *st(w)* and *loc(w)* are numerically different albeit instantiating the same universals.

A simple way to modify L_1 in order to account for coincidence as manifested in Gupta's system can be achieved by:

(a). removing the constraint (7) in definition 4.17;
(b). including the operator ≈ for coincidence with the following semantics: If α is an atomic formula $t_1 \approx t_2$, then $\nu^{w}_{M,a}(\alpha) = T$ if $\nu^{w}_{M,a}(t_1) = \nu^{w}_{M,a}(t_2)$. Otherwise $\nu^{w}_{M,a}(\alpha) = F$;
(c). Defining the identity relation between individual constants as $(t_1 = t_2) =_{def} \Box(t_1 \approx t_2)$, i.e., two continuants are identical if they coincide in every possible world.

Example

The UML specification of figure 4.10 depicts and example of a conceptual specification produced using the profile defined in table 4.1.

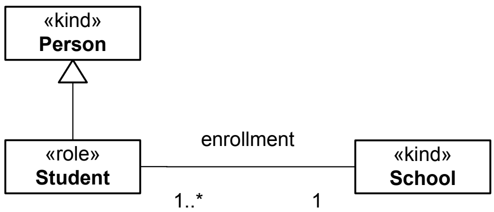


Figure 4-10 Example depicting a phased-sortal role, its allowed type and a relational restriction condition

The L_1 rendering of this specification is presented below:

$\Box(\forall$Student,x person(x))
$\Box(\forall$Student,x $\exists!$School,y enrolled-in(x,y))
$\Box(\forall$School,x $\exists$Student,y enrolled-in(y,x))

According to definition 4.9, $\Box(\forall$Student,x person(x)) is actually an abbreviation for $\Box((\forall$Student,x)($\exists$Person,y) (y = x)).

In this case we have that

K = R = C = {Person}, B = $\varnothing$, A = {Student},
P = {enrolled-in}, T = $\varnothing$

A L_1 model for this specification is presented below:

$\mathcal{W} = \{w,w'\}$
$\mathcal{D}(w) = \{\alpha,\beta,\gamma\}$ $\quad$ $\mathcal{D}(w') = \{\varepsilon,\eta,\iota\}$
$I = \{john, mary, \mathcal{UT}\}$

Individual Concepts

$john(w) = \alpha$ $\quad$ $john(w') = \varepsilon$
$mary(w) = \beta$ $\quad$ $mary(w') = \eta$
$\mathcal{UT}(w) = \gamma$ $\quad$ $\mathcal{UT}(w') = \iota$

Interpretation of Sortals

δ(Person)(*w*) = *{john,mary}* δ(Person)(*w'*) = *{john,mary}*
δ(School)(*w*) = *{UT}* δ(School)(*w'*) = *{UT}*
δ(Student)*(w)* = *{john}* δ(Student)*(w')* = *{mary}*

Interpretation of Predicates

δ(enrolled-in)(*w*) = *{<α,γ>}* δ(enrolled-in)*(w')* = *{<η,ι>}*

4.4.2 Quantifying over substantial individuals

In this section, we consider a modification of language L_1 to arrive at a system L_2 whose elements of quantification are ordinary individuals.

The syntax of L_2 is exactly like that of L_1, i.e., every grammatically valid formula in the latter language is a valid formula of the former, and vice-versa. In L_2 all quantification is still restricted by sortals, and sortals and predicates still differ in the fact that whilst in the extension of the latter we have the primitive objects of quantification, in the extension of the former we have individual concepts.

From a semantic point of view, we make the following modifications in the L_2 in comparison to L_1.

Definition 4.22 (model structure)

A model structure for L_2 is defined as an ordered couple $\langle \mathcal{W}, \mathcal{D} \rangle$ where: (i) $\mathcal{W}$ is a non-empty set of possible worlds; (ii) In L_2, the domain $\mathcal{D}$ of quantification is that of *possibilia*, which includes all possible entities independent of their actual existence. Therefore we shall quantify over a constant domain in all possible worlds. Moreover, all worlds are equally accessible and therefore we omit the accessibility relation from the model structure (Fitting & Mendelsonh, 1998).

■

Given a model structure $\mathcal{M}$ $(= \langle \mathcal{W}, \mathcal{D} \rangle)$, the intention of individual constant can be represented by an individual concept. However, due to the nature of the domain D in L_2, we also have to change the definition of the individual concepts adopted in this language.

Definition 4.23 (individual concept)

Let $\mathcal{M} = \langle \mathcal{W}, \mathcal{D} \rangle$. An individual concept i in $\mathcal{M}$ is function from $\mathcal{W}$ into $\mathcal{D}$.

Let i be an individual concept in $\mathcal{M}$ such that i is a constant function. That is to say: for all $w,w' \in \mathcal{W}$, $i(w) = i(w')$. The individual concepts in $\mathcal{M}$ that have this property are named here L_2-individual concepts.

For a given model structure $\mathcal{M}$ we define I as a set of individual concepts defined for that structure and, $I_C \subseteq I$ as the subset of I that includes only L_2-individual concepts.

■

The intention of n-place predicate and of sortal classifier are defined exactly as in L_1, i.e., as an n-ary *property* and a *sort*, respectively. Notice that the following alternative definition for sorts in L_2 can be given.

Definition 4.24 (sort)

Let $\mathcal{M} = \langle W, D \rangle$ be a L_2 model structure. An *intensional property* in $\mathcal{M}$ is a function ℓ from $\mathcal{W}$ into the powerset of individual concepts in $\mathcal{M}$ (i.e., $\wp(I)$). A sort in $\mathcal{M}$ is an intensional property that has as co-domain the set I_C, i.e., a sort is a function from worlds to sets of L_2-individual concepts.

■

Notice that separation becomes a consequence of this definition for sorts.

<u>Proof</u>: Let ℓ be a sort in L_2, and let i,j be two individual concepts such that there are worlds $w,w' \in \mathcal{W}$, such that $i \in \ell(w)$, and $j \in \ell(w')$. Since i and j are L_2-individual concepts, they are constant functions. Thus, if i and j coincide at a world w'', then they coincide in every world.

□

Again, due to the nature of the domain $\mathcal{D}$ in L_2, we also have to change the definitions of $\ell[w]$ and $\ell|[w]|$.

Definition 4.25

$\ell[w] = \{d$: $d \in \mathcal{D}$ and there is an individual concept $i \in \ell(w)$ such that $i(w) = d\}$.

■

Definition 4.26

$\ell|[w]| = \{d$: $d \in \mathcal{D}$ and there is an individual concept $i \in \ell(w')$ for some $w \in \mathcal{W}$ such that $i(w') = d\}$.

■

The notion of counterpart becomes dispensable in L_2 for reasons that we shall see in the sequel. A model in L_2 is defined as:

Definition 4.27 (model)

A model in L_2 can then be defined as a triple $\langle \mathcal{W}, \mathcal{D}, \delta \rangle$ such that:

1. $\langle \mathcal{W}, \mathcal{D} \rangle$ is a model structure for L_2;
2. δ is an interpretation function assigning values to the non-logical constants of the language such that: it assigns an L_2-individual concept to each individual constant $c \in T$ of L_2; an n-ary property to each n-place predicate $p \in P$ of L_2; a sort to each sortal classifier $S \in C$ of L_2.
3. The interpretation function δ must also satisfy the following the constraints (3) to (7) from definition 4.17.

■

Suppose that $c \in T$ is an individual constant of L_2. According to definition 4.27 above, the interpretation function δ assigns to c a L_2-individual concept i of the model structure $\langle \mathcal{W}, \mathcal{D} \rangle$. Since i is a constant function, we have that the interpretation of an individual constant c (proper name) is world invariant. This amount to Kripke's thesis that proper names are rigid designators (Kripke, 1982) and conforms to Montague's meaning postulate for common nouns 1 (MP1) (Montague, 1974).

Notice that (even without restriction (7) of definition 4.17) statements of relative identity cannot be expressed in L_2. This is because individual constants are always interpreted as L_2-individual concepts. Thus, if two individual concepts i,j that represent the intention of L_2 constants coincide at a world, they coincide in every world (since they are constant functions), even if they do not belong to the same type. Moreover, notice that if $S \in K$ is a kind, and ℓ the sort assigned to S by δ, then we have that $\ell[w] = \ell|[w]|$, for any $w \in \mathcal{W}$. This is due to: (a) since S is a kind then ℓ is a constant function; (b) every individual concept in ℓ is a constant function. In other words, if S is a kind, then every object which is a possible S is actually an S.

In the sequel we define an L_2 assignment.

Definition 4.28 (L_2 assignment)

An assignment for L_2 relative to a model $\langle \mathcal{W}, \mathcal{D}, \delta \rangle$ is a function that assigns to each variable of L_2 an ordered pair $\langle \ell, d \rangle$, where ℓ is a sort relative to the model structure $\langle \mathcal{W}, \mathcal{D} \rangle$ and $d \in \mathcal{D}$. If a is an L_2 assignment then $a_o(x)$

is the object assigned to variable x by a and $a_S(x)$ is the sortal to which x is bound. Moreover, it is always the case that $a_o(x) \in a_S(x)|[w]|$ for all variables.

An L_2 is almost identical to an L_1 assignment, with the difference that in ordered pair $\langle \ell,d \rangle$ assigned to a L_2 variable by a, $d \in$ D.

■

An assignment a' for L_2 is an ℓ variant of a at x is defined exactly as in definition 4.19. Moreover, the notion of a w' variant of an assignment a relative to w (definition 4.20) becomes dispensable, since in L_2 we no longer have the notions of counterpart.

We can then finally define the truth for formulas α in L_2.

Definition 4.29

Finally, let α be an expression in L_2 and, let *the semantic value of* α *at world w in model M and relative to assignment a* be the value of the valuation function $v^{w}_{M,a}$.

With these definitions, we are able to define the semantics of L_2 as follows:

(a). For all cases described in (a) to (g) in definition 4.21, the valuation function in L_2 has exactly the same value as their respectives in L_1.

(b). If α is the formula □A, then $v^{w}_{M,a}(\alpha) = \text{T}$ iff for all $w' \in \mathcal{W}$ then $v^{w'}_{M,a}(\text{A}) = \text{T}$. Otherwise $v^{w}_{M,a}(\alpha) = \text{F}$.

■

The L_1 rendering of the specification of figure 4.10 presented in the previous section is still a syntactically valid L_2 specification. A L_2 model for this specification is presented below:

$\mathcal{W} = \{w,w'\}$
$\mathcal{D} = \{\Delta,\Phi,\Theta\}$
$I = \{john, mary, \mathcal{UT}\}$

Individual Concepts

$john(w) = \Delta$	$john(w') = \Delta$
$mary(w) = \Phi$	$mary(w') = \Phi$
$\mathcal{UT}(w) = \Theta$	$\mathcal{UT}(w') = \Theta$

Interpretation of Sortals

δ(Person)*(w) = {john,mary}* δ(Person)*(w') = {john,mary}*
δ(School)*(w) = {UT}* δ(School)*(w') = {UT}*
δ(Student)*(w) = {john}* δ(Student)*(w') = {mary}*

Interpretation of Predicates

δ(enrolled-in)*(w)* = $\{\langle\Delta,\Theta\rangle\}$ δ(enrolled-in)*(w')* = $\{\langle\Phi,\Theta\rangle\}$

4.5 Related Work

4.5.1 OntoClean

The work presented in this chapter has been influenced by the OntoClean methodology, which proposes a number of methodological guidelines to evaluate the conceptual correctness of specialization relationships (Guarino & Welty, 2004, 2002a, 2002b). Despite bearing a strong similarity with OntoClean's useful property types, the stereotypes presented in table 4.1 also hold some important differences. Firstly, rigid sortals in our typology are always considered to be independent. Examples of rigid sortals that are typically considered dependent are universals whose instances are *features* in the sense of (Gangemi et. al, 2003) (e.g., holes, bumps, stains). In the scope of this work, features are considered parts of their hosts (as opposed to *moments* inhering in them, see chapter 6) and therefore do not qualify as dependent entities according to the definition of Guarino and Welty. The relation between a feature and its host is, thus, considered one of *inseparable parthood*. We say, for instance, that a hole in a piece of cheese is an inseparable part of the cheese, not *externally dependent* on it (see chapter 6). Likewise, substantial individuals can have features as essential parts. For instance, the key whole of a locker may be considered an essential part of the locker. The distinctions between inseparable parts/mandatory wholes and essential/mandatory parts are explained in details in chapter 5 of this thesis. Another consequence of this choice is that *categories*, which typically represent abstractions of different kinds, are also always considered to be independent.

A second difference exists regarding the use of the term *mixin*. In OntoClean, mixin is used to denote a combination (conjunction or disjunction) of rigid and non-rigid properties which are subsumed by at least one sortal. Therefore, in this sense mixin is also a sortal. Another example is a combination of the kind CAT and the formal role WEAPON producing the semi-rigid CAT-or-WEAPON. We believe that the category denoted by this sense of the word mixin, although useful in structuring large ontologies, is of little use in conceptual modeling. Conversely, we use the term mixin here according to its widespread use in the object modeling community, i.e. as an abstract type that can classify instances of different classes but which has no direct instances (Booch, 1994). In this sense, mixins include *category*, *role mixin* but also what is called *attribution* in (Welty & Guarino, 2001). Moreover, the stereotype « roleMixin » used here posses the same meta-properties as the category of *formal roles* in OntoClean. As previously mentioned, role mixins include the category of formal roles but it also includes dispersive types which are abstractions of common properties of material roles.

In OntoClean, the notion of dependence applied to roles is a weaker constraint than the one proposed here. In their case, a role classifier is regarded as *notionally dependent* on another classifier. For instance, the roles *Offspring* and *Parent* are notionally dependent on each other and, thus, for an instance *x* of Offspring to exist another individual *y*, instance of Parent, must also exist. The notion of notional dependence used is the one of (Simons, 1997). Here, this constraint is strengthened: *x* and *y* must also be related via a relationship r, where r is the restriction condition for the two roles.

Finally, in Ontoclean, despite of having the same meta-properties, phases are not explicitly required to be defined as partitions of the supertypes they specialize.

4.5.2 The BWW (Bunge-Wand-Weber) Approach

An approach that shares the same objectives as the work presented here appears in (Evermann & Wand, 2001b; Parsons & Wand, 2000; Wand & Storey & Weber, 1999; Weber, 1997; Wand & Weber, 1989, 1993). In these articles, the authors report their results in mapping common conceptual modeling constructs to an upper level ontology. Their approach is based on the BWW ontology, a framework created by Yair Wand and Ron Weber on the basis of the original metaphysical theory developed by Mario Bunge in (Bunge, 1977). In (Evermann & Wand, 2001b), in particular, the authors propose that a UML class should be used to represent a BWW-natural kind (it should be equivalent to functional schema of a BWW-natural kind). A natural kind is defined by Bunge as *"a set of substantial things that share lawfully related properties"*. According to this definition, the authors' proposal for the interpretation of the UML class meta-construct is equivalent to that of kinds and subkinds (i.e. rigid sortals) defined in this chapter. A law is an essential property of its instances (by definition). Since a natural kind is a grouping of things that share these essential properties it is, also by definition, a rigid class. Equating a natural kind with the denotation of a substance sortal concept of substantials is in conformance with other works in the philosophical literature (Lowe, 1989; van Leeuwen, 1991).

As demonstrated throughout this work, kinds constitute a subset of the category types that are necessary to represent the phenomena available in cognition and language. Therefore, a modeling construct representing a kind is only one of a set of modeling constructs that should be available to the conceptual modeler. For this reason, the typology of classifiers presented in this chapter is not only compatible with the proposals of the BWW approach but extends it towards a much richer system of ontological distinctions.

4.5.3 The Approach of Wieringa, de Jong and Spruit

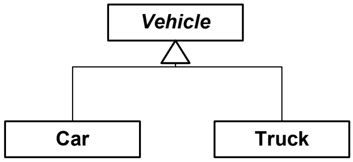
In (Wieringa & de Jong & Spruit, 1995), the authors propose three distinctions in types of taxonomic structures that should be employed to model class migration in object-oriented modeling, namely, *static* and *dynamic subclassing* and *role playing*.

An example of a static taxonomic structure is depicted in figure 4.11. A static structure is always defined as a partition of a superclass that is responsible for supplying a principle of identity for the instances of the subclass. The authors write: *"an instance of a subclass is identical to an instance of a superclass"*. In their approach, the superclass is responsible for generating identifiers for its instances and, hence, for the instances of its subclasses. The authors write that *"object identifiers have a 1-1 relationship with objects and that relation can never be changed"*, i.e., object identifiers are representation of proper names, as here conceived, and also comply with Kripke's requirement of rigidity for proper names. Moreover, instances of the subclass can never migrate to other subclasses in the partition. For instance, a Car can never become a Truck and vice-versa. Since Car and Truck define a partition, ergo, a Car (Truck) can never cease to be a Car (Truck) without ceasing to exist. We can conclude therefore that if we take a *possibilist*[29] view on modality, the extensions of both the superclass and

[29] In a possibilist approach of modal logics, quantifiers range over entities that possibly exist and, therefore, the domain of quantification is constant in every possible world. In actualism, in contrast, quantification considers what actually exist in each world (Fitting & Mendelsonh, 1998).

each of the subclasses in a static partition are *modally constant*. This is, however, not explicitly represented by the authors who take an *actualist* approach. In summary, the superclass and subclasses in a static partition are analogous to what we term here kind and subkind, respectively. Nonetheless, contrary to the authors, we do not require a subkind taxonomic structure to be always defined as a partition of the kind.

Figure 4-11 Example of a static subclassing partition



A *dynamic structure* in their approach is identical to what we term here a *phase partition*. Wieringa and colleagues also explicitly forbid the construction of models in which a static partition is followed upwards in the taxonomic structure by a dynamic partition, i.e., a subclass in a dynamic partition cannot be a superclass in a static one. This is equivalent to say, again if we take a possibilist interpretation of their approach, that a phased-sortal cannot subsume a rigid sortal. The justification for this constraint in Wieringa et al. is that it *enhances the intuitive structure of the models and simplifies their formalization*. Since this constraint is equivalent to our postulate 4.3, the philosophical justification provided for the latter in section 4.1 can also serve as a justification for the former.

One of the main differences between the approach of Wieringa et al. and the one proposed here is w.r.t. the category of role universals. In their approach a role universal is not a phased-sortal. Conversely, roles are rigid classifiers whose instances are said to be *played by* instances of ordinary (static and dynamic) types. The *played by* relation (also termed *inheritance by delegation* by the authors) between a role instance *r* and an object *o* implies that r is *specifically existentially dependent* on *o*. This means that r can only be played by o, and can only exist when being played by *o*. However, in contrast, *o* can possibly be associated via the *play by* relation to many instances of the role class (and to many different role classes). Unlike static and dynamic subtypes, role universals are not required to be defined in a partition. Moreover, role universals are responsible to supplying a principle of identity for its instances, which is different from the one supplied by the universals instantiated by their players.

Figure 4.12 depicts an example of ordinary and a role universal according to the Wieringa, de Jong, and Spruit.

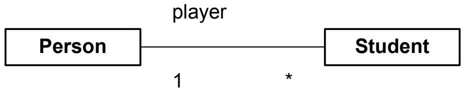


Figure 4-12 Example of Role and Role Player Universals

Although mainly motivated by practical implementation issues, a sound philosophical interpretation can be given to role universals and role individuals as conceived by the authors. In chapter 7, we discuss in depth the relation between the notions of role as proposed by Wieringa, de Jong and Spruit and the one proposed in this chapter. We demonstrate that, albeit strongly related, the two notions of role refer to different ontological entities with incompatible meta-properties. Actually, an even stronger statement can be made: role instances in the two approaches do not even belong to the same meta-level ontological category. Our notion of role universal (as well as the other universals discussed in this chapter) refers to the ontological category of *substantial universals*. Role universals, as conceived by the authors, conversely, refer to universals whose instances are individualized properties, i.e., moment universals (see chapter 6).

We demonstrate throughout this thesis that both notions of role are important for the theory and practice of conceptual modeling. However, we have some remarks about the notion of role as formulated by the authors:

1. In discussing the relations between what they term *roles* and *ordinary objects*, Wieringa and colleagues refer to the philosophical logic relation of *coincidence* of objects (see subsection 4.4.2). For example, a student *s* and a person *p* that plays the role of a student in world w are said to coincide at w. In our perspective, since *s* is actually a moment, the relation between *s* and *p* in this case is a relation of *inherence* (see chapter 6) and not one of coincidence. It is important to emphasize that that coinciding objects are not existentially dependent on each other. For instance, in the statue/lump of clay example previously mentioned, their lifecycles are completely independent but merely coincident at some possible worlds;

2. In the way presented by the authors, role individuals should be interpreted as *intrinsic* individualized properties, since they are existentially dependent on a single individual. For instance, in the aforementioned example, the student *s* is dependent on person *p*. However, this is actually not the case. In fact, student *s* depends not only on person *p* but also on the existence of another individual, say *e*, instance of Educational Institution. Thus, *s* should be conceived as a *relational* individualized property (see chapter 6) not an intrinsic one.

This problem becomes clear in our approach due to our *relational* restriction condition for role universals;

3. Since no restriction is defined for the allowed type of a role universal, optional cardinalities must be represented in the authors approach (see figure 4.12). As argued in, for instance, (Weber, 1997; Wand & Storey & Weber, 1999), from an ontological standpoint, there is no such a thing as an optional property and, hence, the representation of optional cardinality constraints leads to unsound models (in the technical sense of chapter 2) with undesirable consequences in terms of clarity. Furthermore, as empirically demonstrated in (Bodard et al., 2001), conceptual models without optional properties lead to better performance in problem-solving tasks that require a deeper-level understanding of the represented domain.

In summary, our approach is compatible with one proposed by Wieringa and colleagues. However, it also complements their approach by elaborating on the distinctions among the category of non-sortal universals, and by discussing the admissible relations between sortals and non-sortals. Moreover, by also considering the notion of roles as a *substantial sortals*, we extend their approach towards a more complete typology of sortals universals for conceptual modeling.

These issues related to role modeling are comprehensively discussed in chapter 7 of this thesis.

4.6 Final Considerations

In this chapter, we present a well-founded theory of universals for conceptual (ontology) modeling. The theory proposed is founded in a number of results in the literature of philosophy of language and descriptive metaphysics and supported by substantial empirical evidence from research in cognitive psychology.

In section 4.2, this theory is used in the definition of a modeling profile comprised of:

(i) a set of stereotypes representing the ontological distinctions on types of substantial universals proposed by the theory (depicted in figure 4.1);

(ii) Constraints on the possible relations to be established between these elements, representing the postulates of the theory.

In chapter 8, we again make use of the theory proposed here to analyze and extend the UML metamodel, in the main case study of this thesis.

The use of the modeling profile proposed here to address relevant modeling problems in conceptual (ontology) representation in this chapter is only briefly illustrated by examples. In chapter 7 of this thesis, the profile is used in to develop a design pattern that captures a solution to a recurrent real-world problem in *role modeling with disjoint allowed types* discussed in the literature. Additionally, in chapter 8, the ontologically well-founded version of UML which is produced in conformance with the theory presented here is employed to analyze and integrate concurrently developed semantic web (lightweight) ontologies in the scope of a context-aware service platform.

In order to formally characterize the ontological distinctions and postulates proposed by the theory, we have presented two extensions to traditional systems of quantified modal logics, namely, the languages L_1 and L_2 discussed in section 4.4. These languages are far from complete and could be extended in many ways. For example, order-sorted extensions in the lines of (Kaneiwa & Mizoguchi, 2004) or of the dynamic database logic presented in (Wieringa & de Jong & Spruit, 1995) could be proposed. Moreover, in an alternative formulation, the constraints imposed to the interpretation functions of L_1 and L_2 could be directly considered in the object language, as in the sortal logic proposed by (Lowe, 1989) or as in the system proposed by (Freund, 2000). In the latter, not only quantification and identity are restricted by sortal concepts but (second-order) quantification over sortal concepts is also part of the language. The objective here, instead, is only to formally characterize in a simpler way the distinction between rigid and non-rigid sortal universals and the important distinction between sortals and non-sortals w.r.t. to the former's exclusive ability to supply a principle of persistence and transworld identity to its instances.